

Work: $\langle \phi_1 \dots \phi_n \rangle = \frac{1}{Z[g]} \int \mathcal{D}\phi \phi_1 \dots \phi_n e^{-S[\phi, g]}$

$$-S[\phi, g + \delta g] = -S[\phi, g] - \int d^4x \sqrt{g} \frac{\delta S}{\delta g^{\mu\nu}(x)} +$$

$$T_{\mu\nu} = -\frac{2}{\sqrt{g}} \frac{\delta S}{\delta g^{\mu\nu}}$$

$$\langle \phi_1 \dots \phi_n \rangle_{g+\delta g} = \frac{1}{Z[g+\delta g]} \int \mathcal{D}\phi \phi_1 \dots \phi_n e^{-S[\phi, g]} \left(1 + \frac{1}{2} \int d^4x \sqrt{g} \delta g^{\mu\nu} T_{\mu\nu} + \dots \right)$$

$$= \frac{1}{Z[g+\delta g]} \left(\int \mathcal{D}\phi \phi_1 \dots \phi_n e^{-S[\phi, g]} + \frac{1}{2} \int d^4x \sqrt{g} \delta g^{\mu\nu} \langle \phi_1 \dots \phi_n T_{\mu\nu} \rangle_g \right)$$

$$= \frac{1}{Z[g+\delta g]} \left(Z[g] + \frac{1}{2} \int d^4x \sqrt{g} \delta g^{\mu\nu} \langle T_{\mu\nu} \phi_1 \dots \phi_n \rangle_g \right) Z[g] \dots$$

$$Z[g+\delta g] = \int \mathcal{D}\phi e^{-S[\phi, g]} e^{+\frac{1}{2} \int d^4x \sqrt{g} \delta g^{\mu\nu} T_{\mu\nu}}$$

$$= \int \mathcal{D}\phi e^{-S} \left(1 + \frac{1}{2} \int d^4x \sqrt{g} \delta g^{\mu\nu} T_{\mu\nu} + \dots \right)$$

$$= Z[g] + \frac{1}{2} \int d^4x \sqrt{g} \delta g^{\mu\nu} \langle T_{\mu\nu} \rangle_g e^{-S}$$

$$\frac{Z[g+\delta g]}{Z[g]} = 1 + \frac{1}{2} \int d^4x \sqrt{g} \delta g^{\mu\nu} \langle T_{\mu\nu} \rangle_g$$

$$\langle \phi_1 \dots \phi_n \rangle_{g+\delta g} = \frac{1}{(1 + \frac{1}{2} \langle T \rangle)} \left(\langle \phi_1 \dots \phi_n \rangle_g + \frac{1}{2} \int d^4x \sqrt{g} \delta g^{\mu\nu} \langle \phi_1 \dots \phi_n T_{\mu\nu} \rangle_g \right)$$

$$= \left(1 - \frac{1}{2} \int d^4x \sqrt{g} \delta g^{\mu\nu} \langle T_{\mu\nu} \rangle_g \right) \left(\langle \phi_1 \dots \phi_n \rangle_g + \frac{1}{2} \int d^4x \sqrt{g} \delta g^{\mu\nu} \langle \phi_1 \dots \phi_n T_{\mu\nu} \rangle_g \right)$$

$$= \langle \phi_1 \dots \phi_n \rangle_g + \frac{1}{2} \int d^4x \sqrt{g} \delta g^{\mu\nu} \langle \phi_1 \dots \phi_n T_{\mu\nu} \rangle_g - \dots$$

$$\langle \phi_1 \dots \phi_n \rangle_{g+\delta g} - \langle \phi_1 \dots \phi_n \rangle_g =$$

$$\frac{1}{2} \int d^4x \sqrt{g} \delta g^{\mu\nu} \left(\langle T_{\mu\nu} \phi_1 \dots \phi_n \rangle_g - \langle T_{\mu\nu} \rangle_g \langle \phi_1 \dots \phi_n \rangle_g \right)$$

$$\delta \langle \phi \rangle = \int \mathcal{D}\phi (\delta \phi e^{-S})$$

$$= \langle \delta \phi \rangle + \int \mathcal{D}\phi \phi \delta S e^{-S}$$

$$= \langle \delta x \rangle + \int \delta A \lambda \delta A \frac{\delta \mathcal{L}}{\delta A} e$$

$$= \langle \delta x \rangle + \int \delta A \delta \mathcal{L}$$